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Original Article

Retrospective Chart Review of Euglycemic Diabetic Ketoacidosis Rates and Outcomes Postimplementation of Sodium Glucose Cotransporter 2 Inhibitor Use Stoppage 5 Days Before Open Heart Surgery

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Objectives: Outcomes of stopping sodium-glucose cotransporter 2 inhibitors (SGLT2i) 5 days before open heart surgery (euglycemic diabetic ketoacidosis [eDKA] rate, mortality, infection, hospital, and cardiovascular intensive care unit [CVICU] length of stay [LOS]).
Design: Retrospective study.
Setting: Academic university hospital.
Participants: Adult open heart surgery patients.
Exposures: Patients on SGLT2i who stopped medications 5 days before open heart surgery versus non−SGLT2i-using patients.
Measurements and Main Results: We evaluated patients who were told to stop SGLT2i 5 days before receiving open heart surgery for eDKA development within 24 hours of hospital admission (12/14/2022 to 10/25/2023). Non-eDKA outcomes were compared between SGLT2i users and non-users using Wilcoxon rank sum and Chi-square testing as appropriate. Of 540 open heart surgery patients, 48 (8.9%) were prescribed SGLT2i's before surgery; of these, 0 (0%) developed eDKA within 24 hours of hospital admission. Hospital LOS was statistically longer for patients with SGLT2i use (median [interquartile range]: 4 [4-5] *v* 4 [3-6] days, p = 0.003; mean [standard deviation]: 4 [3] *v* 5 [2], p = 0.03). We found no significant difference between patients with and without SGLT2i use in CVICU LOS (median [interquartile range]: 1.67 [0.95-2.09] *v* 1.17 [0.96-1.88] days, p = 0.14), in-hospital mortality (2.1% [1] *v* 0.6% [3], p = 0.3), or sternal infections (0 [0.0%] *v* 2 [0.4%], p > 0.9).
Conclusion: Postoperative eDKA was absent in SGLT2i patients told to stop medications 5 days before open heart surgery. LOS and infection and mortality rates appeared similar between the two cohorts. Stopping SGLT2i medications 5 days before open heart surgery appears safe.
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Key Words: cardiothoracic surgery; euglycemic; diabetic ketoacidosis; metabolic acidosis; length of stay; mortality; infections

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DIABETES MELLITUS (DM) is a common condition affecting approximately 12% of the US adult population and 25% of the older population.[1] The incidence of DM is increasing.[2] Type 2 DM is more common[3] and is due to peripheral insulin resistance and/or decreased insulin secretion.[4] One of the many therapies for type 2 DM is sodium glucose cotransporter 2 inhibitors (SGLT2i). These medications selectively inhibit SGLT2 in the kidneys, leading to glucosuria. There are numerous SGLT2i on the market (Table 1). There are many benefits to SGLT2i. They have been shown to be nephroprotective, reduce blood pressure, and provide cardiovascular protection.[5-10] Most importantly, in four large trials they have been shown to decrease heart failure exacerbations and all-cause mortality.[11-14] Based on these data, large populations of diabetic patients at increased risk for heart failure are being started on these medications.[15] There are many side effects of SGLT2i that have been reported. One of the more serious side effects described is euglycemic diabetic ketoacidosis (eDKA). eDKA is typically described in the literature as meeting DKA criteria (increased anion gap metabolic acidosis from ketosis) in the setting of euglycemia.[16,17] However, we were unable to find a gold standard for the diagnosis of eDKA specifically.

The Food and Drug Administration (FDA) recommends stopping SGLT2i 3 to 4 days before surgery due to the increased risk of eDKA.[18] We were unable to find robust data from real-world clinical experience on when to appropriately stop these medications, especially in the pre−cardiac surgery period. Guidelines from the American Association of Clinical Endocrinologists and the American College of Endocrinology recommend stopping SGLT2i at least 24 hours before surgery.[19] Pharmacodynamic effects can persist due to the prolonged half-life of these agents (11-17 hours),[20] putting patients at risk for postoperative eDKA despite stopping these agents preoperatively as per guideline recommendations. There is literature that suggests stopping SGLT2i at least 5 days before surgery may be beneficial.[21] There is also concern that patients with renal failure may need SGLT2i to be stopped even earlier as these medications are renally cleared.[22,23]

With these data in mind, on December 14, 2022, the surgical department at our institution began notifying patients to stop taking any SGLT2i medications 5 days before planned open heart surgery. This decision was based on surgeon preference after review of the FDA recommendation (of holding for 3-4 days) and case reports. We sought, first, to investigate the frequency of eDKA in our postimplementation phase. Additionally, we compared the clinical outcomes of these patients to a cohort never on SGLT2i. Finally, mortality rates and sternal wound infection rates are presented.

We hypothesized that our intervention would reduce the rates of eDKA postoperatively compared to our historical rates based on prior research[17] and that the outcomes of these patients postoperatively would not be worse than those never on an SGLT2i.

Methods

We performed a retrospective chart review on all post−open heart surgery patients who were operated on within 24 hours of admission at our academic hospital over an approximately 10-month period (December 14, 2022, to October 25, 2023), post-implementation of our policy recommending patients discontinue SGLT2i use 5 days before surgery). Of note, we did not have any other bridging techniques in our patient's preoperative hyperglycemia medications. Patients were specifically told to hold the SGLT2i for 5 days before surgery. Patients under 18 years of age were excluded. Because we wanted to study a homogenous post−heart surgery cohort, we excluded patients who were operated on after 24 hours following hospital admission. Institutional review board approval was obtained with a waiver of informed consent.

Variables considered were (1) the use of SGLT2i before surgery and (2) the development of DKA. Preoperative SGLT2i use was established by automated data extraction from our electronic health record (Epic) and enhanced manual chart review for hospitalizations including Current Procedural Terminology (CPT) codes for open heart surgery diagnosis and

Table 1
List of SGLT2i Medications Available in the United States by Generic and Trade Names

SGLT2i Generic Name	SGLT2i Trade Name	Combination SGLT2i Generic Name	Combination SGLT2i Trade Name
Bexagliflozin	Brenzavvy	Canagliflozin and metformin	Invokamet
Canagliflozin	Invokana	Canagliflozin and metformin XR	Invokamet XR
Dapagliflozin	Farxiga	Dapagliflozin and metformin XR	Xigduo XR
Empagliflozin	Jardiance	Dapagliflozin and saxagliptin	Qtern
Ertugliflozin	Steglatro	Empagliflozin and linagliptin	Glyxambi
Sotagliflozin	Inpefa	Empagliflozin and metformin	Synjardy
		Empagliflozin and metformin XR	Synjardy XR
		Ertugliflozin and metformin	Segluromet
		Ertugliflozin and sitagliptin	Steglujan
		Dapagliflozin, saxagliptin, and metformin hcl XR	Qternment XR
		Empagliflozin, linagliptin, and metformin hcl XR	Trijardy XR

Table 2
Demographics and Clinical Characteristics Stratified by SGLT2i Use

Characteristic	Overall, N = 540[1]	No, N = 492[1]	Yes, N = 48[1]	p-Value[2]
Age (years)	67 (60, 74)	67 (60, 74)	68 (60, 72)	>0.9
Female sex	189 (35%)	182 (37%)	7 (15%)	**0.002**
Race/ethnicity				0.2
Hispanic Black	3 (0.6%)	2 (0.4%)	1 (2.1%)	
Hispanic White	262 (49%)	234 (48%)	28 (58%)	
Non-Hispanic Black	21 (3.9%)	19 (3.9%)	2 (4.2%)	
Non-Hispanic White	212 (39%)	198 (40%)	14 (29%)	
Other/unknown	42 (7.8%)	39 (7.9%)	3 (6.3%)	
BMI	27.7 (24.8, 31.5)	27.4 (24.7, 31.2)	30.8 (27.1, 33.2)	**0.001**
Obesity	182 (34%)	156 (32%)	26 (54%)	**0.002**
Anemia	123 (23%)	109 (23%)	14 (29%)	0.3
Valve disease	411 (78%)	383 (79%)	28 (58%)	<**0.001**
CPD	4 (0.8%)	3 (0.6%)	1 (2.1%)	0.3
Payor				0.7
Commercial	241 (45%)	220 (45%)	21 (44%)	
Medicaid	14 (2.6%)	13 (2.6%)	1 (2.1%)	
Medicare	273 (51%)	249 (51%)	24 (50%)	
Other	12 (2.2%)	10 (2.0%)	2 (4.2%)	
Chronic liver disease	9 (1.7%)	6 (1.2%)	3 (6.3%)	**0.039**
Infective endocarditis	15 (2.8%)	15 (3.1%)	0 (0%)	0.4
Creatinine >2 mg/dL	12 (2.3%)	10 (2.1%)	2 (4.2%)	0.3
Preexisting dialysis	8 (1.5%)	8 (1.7%)	0 (0%)	>0.9
EF <40	82 (15%)	64 (13%)	18 (38%)	<**0.001**
Pulmonary hypertension	16 (3.0%)	14 (2.9%)	2 (4.2%)	0.6
Heart failure	111 (21%)	95 (20%)	16 (33%)	**0.026**
Hypertension	421 (79%)	376 (78%)	45 (94%)	**0.009**
Diabetes	127 (24%)	91 (19%)	36 (75%)	<**0.001**
Preoperative insulin use	33 (6.1%)	20 (4.1%)	13 (27%)	<**0.001**
Surgery				
AVR	94 (17%)	88 (18%)	6 (13%)	
CABG	132 (24%)	108 (22%)	24 (50%)	
Double valve	71 (13%)	64 (13%)	7 (15%)	
MVR	153 (28%)	149 (30%)	4 (8.3%)	
Root/ascending aorta	57 (11%)	52 (11%)	5 (10%)	
TVR	10 (1.9%)	9 (1.8%)	1 (2.1%)	
Other	23 (4.3%)	22 (4.5%)	1 (2.1%)	
Previous cardiac surgery	53 (10.0%)	47 (9.7%)	6 (13%)	0.6
Elective	514 (97%)	470 (98%)	44 (92%)	**0.048**
Min CO_2	21.00 (20.00, 22.00)	21.00 (20.00, 22.00)	20.00 (19.00, 21.00)	<**0.001**
Min arterial pH	7.32 (7.30, 7.35)	7.32 (7.30, 7.35)	7.31 (7.28, 7.33)	**0.004**
Min venous pH	7.30 (7.28, 7.33)	7.30 (7.28, 7.33)	7.28 (7.25, 7.31)	<**0.001**

1 Median (IQR); n (%).
2 Wilcoxon rank sum test; Pearson chi-squared test; Fisher exact test.
Abbreviations: AVR, aortic valve replacement; BMI, body mass index; CABG, coronary artery bypass grafting; CPD, chronic pulmonary disease; EF, ejection fraction; MVR, mitral valve replacement; Min arterial pH, minimum arterial pH; Min CO_2, minimum carbon dioxide; Min venous pH, minimum venous pH; TVR, tricuspid valve replacement.

cardiac surgeons at our institution. eDKA diagnosis was made among patients on SGLT2i before surgery if they had a serum bicarbonate less than 18 mmol/L or an arterial or venous blood pH less than 7.3 within 24 hours postsurgery, dextrose levels less than 252 mg/dL, and ketones in the serum or urine.[16,17]

The primary studied outcome was the eDKA rate among patients on SGLT2i preoperatively after the implementation of a policy to instruct them to discontinue these medications 5 days before surgery. Secondary outcomes included cardiovascular intensive care unit (CVICU) length of stay (LOS) and hospital LOS, hospital mortality, sternal wound infections, glycemic control, and any Society of Thoracic Surgeons (STS)–reported infection (sepsis, pneumonia, and peripheral cannulation site infections).[24]

Standard summary statistics were used to describe the cohort, with comparisons between SGLT2i users and non-users made using chi-squared and Wilcoxon rank-sum testing as appropriate. Values of $p < 0.05$ were considered to be

Table 3
Clinical Outcomes SGLT2i Versus Non-SGLT2i

Characteristic	Overall N = 540[1]	No N = 492[1]	Yes N = 48[1]	p-Value[2]
Metabolic acidosis with increased anion gap	2 (0.4%)	2 (0.4%)	0 (0%)	>0.9
Glycemic control (daily average mg/dL)	106 (96, 125)	104 (95, 122)	134 (110, 169)	<0.001
In-hospital mortality	4 (0.8%)	3 (0.6%)	1 (2.1%)	0.3
Permanent stroke	2 (0.4%)	1 (0.2%)	1 (2.1%)	0.2
Blood transfusion	139 (26%)	126 (26%)	13 (27%)	0.9
Postoperative pneumonia	4 (0.8%)	4 (0.8%)	0 (0%)	>0.9
Postoperative sepsis	1 (0.2%)	1 (0.2%)	0 (0%)	>0.9
Surgical site infection	3 (0.6%)	3 (0.6%)	0 (0%)	>0.9
Thoracotomy site infection (postoperative)				
0	530 (100%)	482 (100%)	48 (100%)	
Harvest site infection	1 (0.2%)	1 (0.2%)	0 (0%)	>0.9
Cannulation site infection				
0	530 (100%)	482 (100%)	48 (100%)	
Sternal wound infection (superficial or deep)	2 (0.4%)	2 (0.4%)	0 (0%)	>0.9
Superficial sternal wound infection	1 (0.2%)	1 (0.2%)	0 (0%)	>0.9
Deep sternal wound infection	1 (0.2%)	1 (0.2%)	0 (0%)	>0.9
Non-infectious surgical wound dehiscence				
0	530 (100%)	482 (100%)	48 (100%)	
RBC units				>0.9
0	391 (74%)	356 (74%)	35 (73%)	
1-2	104 (20%)	93 (19%)	11 (23%)	
3-4	27 (5.1%)	25 (5.2%)	2 (4.2%)	
5+	8 (1.5%)	8 (1.7%)	0 (0%)	
FFP units				0.2
0	491 (93%)	447 (93%)	44 (92%)	
1-2	26 (4.9%)	25 (5.2%)	1 (2.1%)	
3-4	9 (1.7%)	7 (1.5%)	2 (4.2%)	
5+	4 (0.8%)	3 (0.6%)	1 (2.1%)	
Platelet units				0.11
0	499 (94%)	456 (95%)	43 (90%)	
1-2	21 (4.0%)	17 (3.5%)	4 (8.3%)	
3-4	7 (1.3%)	7 (1.5%)	0 (0%)	
5+	3 (0.6%)	2 (0.4%)	1 (2.1%)	
Cryo units				0.3
0	526 (99%)	479 (99%)	47 (98%)	
1-2	4 (0.8%)	3 (0.6%)	1 (2.1%)	
Perioperative Afib	117 (22%)	110 (23%)	7 (15%)	0.2
Ventilation time (hour)	3.6 (2.6, 5.0)	3.4 (2.6, 4.8)	4.4 (3.2, 7.6)	0.002
Prolonged ventilation (>24 hours)	16 (3.0%)	13 (2.7%)	3 (6.3%)	0.2
ICU time (hour)	27 (23, 45)	27 (23, 45)	37 (22, 51)	0.2
Length of stay (median, days)	4 (3, 5)	4 (3, 5)	4 (4, 6)	0.003
Length of stay (mean, days)	4 (3)	4 (3)	5 (2)	0.003

1 n (%); median (Q1, Q3).
2 Fisher exact test; Wilcoxon rank sum test; Pearson chi-squared test.
Abbreviations: RBC, red blood cell; FFP, fresh frozen plasma; cryo, cryoprecipitate; Afib, atrial fibrillation; ICU, intensive care unit.

statistically significant. To evaluate the impact of discontinuing SGLT2i before surgery on clinical outcomes, propensity score matching (PSM) using the MatchIt package in R version 4.4.1 (June 14, 2024, "Race for Your Life") was applied with nearest-neighbor matching, a 2:1 control-to-treatment ratio, and a 0.1 caliper and adjustment for sex, body mass index, insulin use, anemia, valve disease, liver disease, infective endocarditis, dialysis, low ejection fraction, heart failure, hypertension, diabetes, surgical procedure type, and elective status (selected based on clinical relevance and statistical significance in baseline comparisons).

Results

Cohort Characteristics and eDKA Frequency

A total of 540 patients had open heart surgery over the studied time period (December 14, 2022, to October 25, 2023), of whom 48 (8.9) were on an SGLT2i. Of the 48 SGLT2i users, 0 (0%) were diagnosed with eDKA. SGLT2i use was associated with increased comorbid illness including higher body mass index, obesity rates, chronic liver disease, low baseline ejection fraction, hypertension, and diabetes (Table 2).

Table 4
Glycemic Outcomes

Characteristic	Overall, N = 540[1]	No, N = 492[1]	Yes, N = 48[1]	p-Value[2]
Hypoglycemia (FS < 70) after 24 hours	7 (1.3%)	6 (1.2%)	1 (2.1%)	0.5
Unknown	12	11	1	
Hypoglycemia (FS < 70) first 24 hours	4 (0.7%)	4 (0.8%)	0 (0%)	>0.9
Hyperglycemia (FS > 180) after 24 hours	139 (26%)	109 (23%)	30 (64%)	**<0.001**
Unknown	12	11	1	
Hyperglycemia (FS > 180) first 24 hours	206 (38%)	174 (35%)	32 (67%)	**<0.001**
Outside ideal range (140-180 after 24 hours	307 (58%)	270 (56%)	37 (79%)	**0.003**
Unknown	12	11	1	
Outside ideal range (140-180) first 24 hours	502 (93%)	457 (93%)	45 (94%)	>0.9
Hypoglycemia episodes (FS < 70) after 24 hours				0.2
0	511 (97%)	467 (97%)	44 (94%)	
1-2	13 (2.5%)	11 (2.3%)	2 (4.3%)	
3-5	3 (0.6%)	2 (0.4%)	1 (2.1%)	
6-10	1 (0.2%)	1 (0.2%)	0 (0%)	
>10	0 (0%)	0 (0%)	0 (0%)	
Unknown	12	11	1	
Hypoglycemia episodes (FS < 70): first 24 hours				0.7
0	523 (97%)	475 (97%)	48 (100%)	
1	13 (2.4%)	13 (2.6%)	0 (0%)	
2	4 (0.7%)	4 (0.8%)	0 (0%)	
Hyperglycemia episodes (FS > 180): after 24 hours				**<0.001**
0	317 (60%)	304 (63%)	13 (28%)	
1-2	101 (19%)	94 (20%)	7 (15%)	
3-5	37 (7.0%)	28 (5.8%)	9 (19%)	
6-10	43 (8.1%)	33 (6.9%)	10 (21%)	
>10	30 (5.7%)	22 (4.6%)	8 (17%)	
Unknown	12	11	1	
Hyperglycemia episodes (FS > 180): first 24 hours				**<0.001**
0	256 (47%)	244 (50%)	12 (25%)	
1-2	140 (26%)	130 (26%)	10 (21%)	
3-5	80 (15%)	69 (14%)	11 (23%)	
6-10	48 (8.9%)	36 (7.3%)	12 (25%)	
>10	16 (3.0%)	13 (2.6%)	3 (6.3%)	
Outside range episodes (140-180): after 24 hours				**<0.001**
0	115 (22%)	114 (24%)	1 (2.1%)	
1-2	184 (35%)	169 (35%)	15 (32%)	
3-5	130 (25%)	117 (24%)	13 (28%)	
6-10	73 (14%)	60 (12%)	13 (28%)	
>10	26 (4.9%)	21 (4.4%)	5 (11%)	
Unknown	12	11	1	
Outside range episodes (140-180): first 24 hours				0.12
0	23 (4.3%)	22 (4.5%)	1 (2.1%)	
1-2	40 (7.4%)	38 (7.7%)	2 (4.2%)	
3-5	155 (29%)	146 (30%)	9 (19%)	
6-10	198 (37%)	180 (37%)	18 (38%)	
>10	124 (23%)	106 (22%)	18 (38%)	

1 n (%).
2 Fisher exact test; Pearson chi-squared test.

Outcomes Comparison between SGLT2i Users and Non-Users

Hospital LOS was statistically significantly longer for patients with SGLT2i use (median [interquartile range]: 4 [4,5] *v* 4 [3,**6**] days, p = 0.003; mean [standard deviation]: 4 [3] *v* 5 [2], p = 0.03). There was no significant difference in CVICU LOS between patients prescribed an SGLT2i and those who were not (1.67 [0.95-2.09] *v* 1.17 [0.96-1.88] days, p = 0.14). Surgical site infection (prior SGLT2i use *v* not: 0 (0.0%) *v* 3 (0.6%), p > 0.9) and in-hospital mortality (prior SGLT2i use *v* not: 1 (2.1%) *v* 3 (0.6%), p = 0.3) were similarly rare in both cohorts (Table 3). There were increased rates of hyperglycemia in the SGLT2i patients versus non-SGLT2i patients both in the first 24 hours and in the subsequent time the patient remained in the hospital (Table 4, Fig 2).

Before matching, variables such as BMI (standardized mean difference [SMD] = 0.46) and insulin use (SMD = 0.52) showed imbalances. After matching, the effective sample size

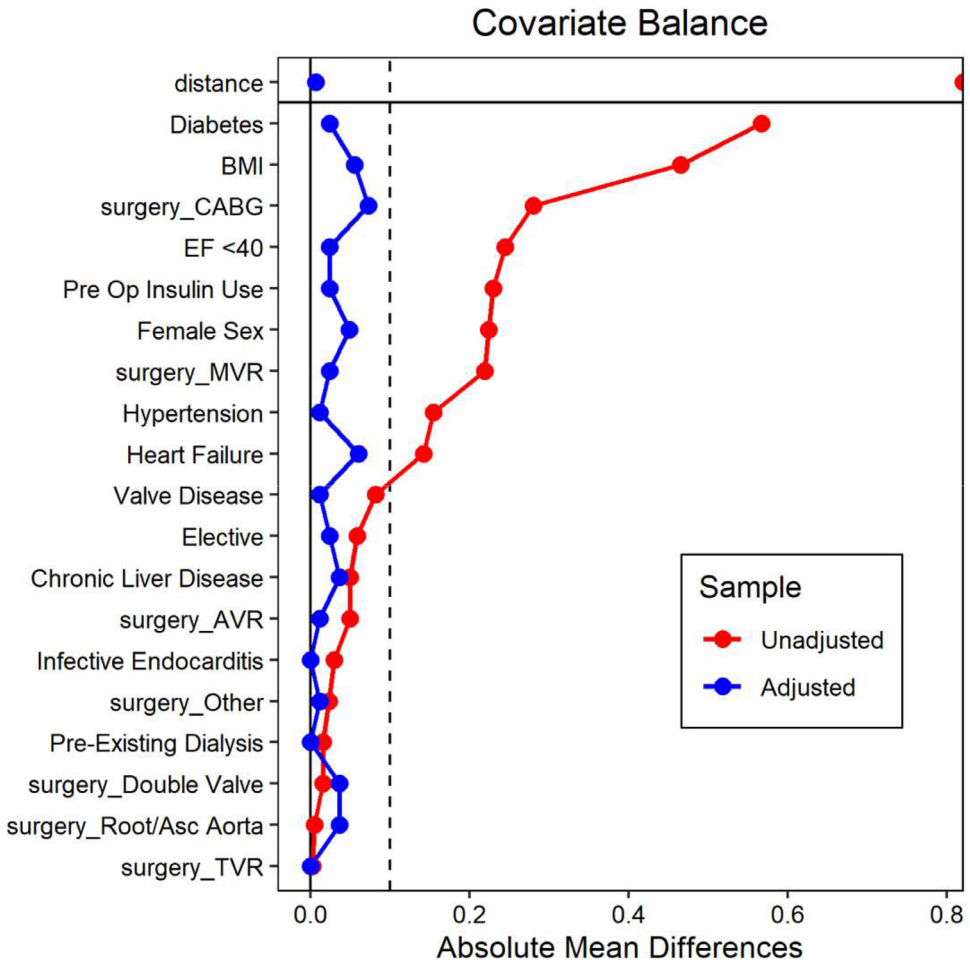


Fig 1. Covariate balance.



Fig 2. Glycemic control during hospital stay stratified by SGLT2i use.

(ESS) for the matched cohort was 69 controls and 41 treated patients, with only 7 treated patients unmatched. All SMDs were reduced below 0.2, indicating adequate balance across covariates (Fig 1, Table 5).

After propensity score matching, there were 110 patients (n = 41 SGLT2i users, 69 control non-users). The matched cohorts had comparable outcomes, with no significant differences in ICU time (median [interquartile range]: 33 [22,47] *v* 26 [21,43] hours, p = 0.2) or hospital LOS (4 [4,5] *v* 4 [3,5] days, p = 0.2), mortality (1% (2.4%) *v* 1% (1.4%), p > 0.9), or surgical site infection rates (0% (0%) *v* 2% (2.9%), p = 0.5) between patients who discontinued SGLT2i and non-users (Table 6).

Table 5
Matched Demographics and Clinical Characteristics Stratified By SGLT2i Use:

Characteristic	No, n = 69[1]	Yes, n = 41[1]	Difference[2]
Age (years)	69 (62, 75)	68 (60, 72)	0.14
Female sex	9 (13%)	7 (17%)	−0.11
Race/ethnicity			0.24
Hispanic White	33 (48%)	24 (59%)	
Non-Hispanic Black	6 (8.7%)	2 (4.9%)	
Non-Hispanic White	23 (33%)	12 (29%)	
Other/unknown	7 (10%)	3 (7.3%)	
BMI	28.5 (26.1, 32.9)	30.2 (26.9, 32.6)	−0.14
Obesity	29 (42%)	21 (51%)	−0.19
Anemia	23 (33%)	10 (24%)	0.20
CPD	0 (0%)	1 (2.4%)	−0.22
Payor			0.31
Commercial	24 (35%)	18 (44%)	
Medicaid	1 (1.4%)	1 (2.4%)	
Medicare	43 (62%)	20 (49%)	
Other	1 (1.4%)	2 (4.9%)	
Chronic liver disease	2 (2.9%)	3 (7.3%)	−0.20
Infective endocarditis	0 (0%)	0 (0%)	0.00
Creatinine >2 mg/dL	1 (1.4%)	1 (2.4%)	−0.07
Preexisting dialysis	0 (0%)	0 (0%)	0.00
EF <40	24 (35%)	13 (32%)	0.07
Pulmonary hypertension	1 (1.4%)	2 (4.9%)	−0.20
Heart failure	24 (35%)	13 (32%)	0.07
Hypertension	64 (93%)	38 (93%)	0.00
Diabetes	44 (64%)	29 (71%)	−0.15
Preoperative insulin use	11 (16%)	9 (22%)	−0.15
Surgery			0.24
AVR	11 (16%)	6 (15%)	
CABG	26 (38%)	19 (46%)	
Double valve	12 (17%)	5 (12%)	
MVR	7 (10%)	4 (9.8%)	
Root/ascending aorta	11 (16%)	5 (12%)	
TVR	1 (1.4%)	1 (2.4%)	
Other	1 (1.4%)	1 (2.4%)	
Previous cardiac surgery	6 (8.7%)	6 (15%)	−0.19
Elective	66 (96%)	38 (93%)	0.13

1 Median (IQR); n (%).
2 Standardized mean difference.
Abbreviations: AVR, aortic valve replacement; BMI, body mass index; CABG, coronary artery bypass grafting; CPD, chronic pulmonary disease; EF, ejection fraction; MVR, mitral valve replacement; TVR, tricuspid valve replacement.

Discussion

We provide the only study to our knowledge looking at rates of eDKA postimplementation of an across-the-board recommendation to stop SGLT2i 5 days before open heart surgery. There is limited literature on eDKA outcomes with shorter duration stoppage (<3 days).[25,26] This literature, however, is largely based on case reports and mixed surgical patient populations.[18,20-23] We implemented this policy after our prior study revealed that 15.1% of SGLT2i users were developing eDKA.[17] Of note, these patients were not told by our facility to specifically stop these medications before surgery. In this current study, we found the rate of eDKA was 0%. Moreover, we did not find any significant differences in clinical outcomes between SGLT2i users instructed in this way and non-SGLT2i users. We did, however, find a statistically significant difference between patients with and without SGLT2i use in hospital LOS, although this did not hold true in our matched cohort. We also found increased rates of hyperglycemia in our SGLT2i patients, although this did not seem to translate to worse clinical outcomes. This is not surprising as SGLT2i patients were not on their baseline glycemic medication and there were differences in baseline diabetes rates between groups. Interestingly, as we hypothesized in the past,[17] the rates of SGLT2i use in our cardiac surgical patients have significantly increased from 3.2% over an approximately 3-year period in the past to 8.9% over approximately 10 months. Based on guideline recommendations and data showing improved patient outcomes with SGLT2i use, we hypothesize that this rate will continue to increase, stressing the importance of proper management of these patients preoperatively to decrease postoperative complications.

The rates of eDKA in SGLT2i-consuming post cardiac surgery patients has been described in the literature as anywhere between 15.1 and 70.8%.[17,27-30] Among non−cardiac surgery patients, the literature is sparser and based on case reports.[31-33] The true risk in this patient population is yet to be described. It would appear that although eDKA has historically thought to be a rare complication in SGLT2i patients, it is likely underreported and may be common in patient groups that are not stopping the SGLT2i in a timely fashion before surgery.

Based on the scant literature we were able to find, SGLT2i medications have been shown to reduce morbidity and mortality in open heart surgery patients and interventional cardiology patients.[34-35] In non-surgical patients, SGLT2i medications have been associated with numerous benefits, including cardiovascular protection, renal protection, decreased exacerbation of heart failure, and all-cause mortality.[36-45] These medications have therefore become guideline medications in heart failure patients.[46]

We also performed propensity score matching to match patient baseline characteristics. Although we found statistically significant increases in glycemic levels between these groups (increased in the SGLT2i group), the differences were likely clinically insignificant.

Table 6
Matched Results for Clinical Outcomes

Characteristic	Overall, N = 110[1]	No, n = 69[1]	Yes, n = 41[1]	p-Value[2]
Metabolic acidosis with increased anion gap	0 (0%)	0 (0%)	0 (0%)	
Glycemic control (daily average mg/dL)	131 (108, 157)	128 (105, 155)	141 (114, 169)	0.3
In-hospital mortality	2 (1.8%)	1 (1.4%)	1 (2.4%)	>0.9
Permanent stroke	1 (0.9%)	0 (0%)	1 (2.4%)	0.4
Blood transfusion	30 (27%)	20 (29%)	10 (24%)	0.6
Postoperative pneumonia	0 (0%)	0 (0%)	0 (0%)	
Postoperative sepsis	0 (0%)	0 (0%)	0 (0%)	
Surgical site infection	2 (1.8%)	2 (2.9%)	0 (0%)	0.5
Thoracotomy site infection (postoperative)	0 (0%)	0 (0%)	0 (0%)	
Harvest site infection	1 (0.9%)	1 (1.4%)	0 (0%)	>0.9
Cannulation site infection	0 (0%)	0 (0%)	0 (0%)	
Sternal wound infection (superficial or deep)	1 (0.9%)	1 (1.4%)	0 (0%)	>0.9
Superficial sternal wound infection	0 (0%)	0 (0%)	0 (0%)	
Deep sternal wound infection	1 (0.9%)	1 (1.4%)	0 (0%)	>0.9
Non-infectious surgical wound dehiscence	0 (0%)	0 (0%)	0 (0%)	
RBC units				0.5
0	80 (73%)	49 (71%)	31 (76%)	
1-2	25 (23%)	17 (25%)	8 (20%)	
3-4	3 (2.7%)	1 (1.4%)	2 (4.9%)	
5+	2 (1.8%)	2 (2.9%)	0 (0%)	
FFP units				0.10
0	105 (95%)	68 (99%)	37 (90%)	
1-2	2 (1.8%)	1 (1.4%)	1 (2.4%)	
3-4	2 (1.8%)	0 (0%)	2 (4.9%)	
5+	1 (0.9%)	0 (0%)	1 (2.4%)	
Platelet units				0.10
0	103 (94%)	67 (97%)	36 (88%)	

This study has several limitations. Due to the retrospective nature of our study, we may have missed SGLT2i patients if medication use was not properly documented on the medical record (chart review was used to find SGLT2i patients). Two such patients who had metabolic acidosis, ketosis, and euglycemia were identified but were not documented to be on an SGLT2i in the medical record; one of these patients was a type 1 diabetic and the other one had no history of diabetes. Whether they truly represented additional cases of eDKA in the setting of undocumented SGLT2i use is unknown. Of note, in both cases there was no prolonged insulin infusion use, and their acidosis resolved. Additionally, we did not make a direct comparison of eDKA rates in our study when compared to prior studies. Instead, we compared this study's eDKA rate to historical rates.

Similarly, we do not know why patients were prescribed an SGLT2i (diabetes management, renal protection, and/or cardiovascular protection); eDKA risk may differ across these groups and may account for the differences between our study and other studies in the literature. Moreover, as a single-center study, the generalizability of our findings to other centers is unknown. And, as in any retrospective study, even after propensity score matching, the potential for residual confounding remains. Finally, although we did not see worse clinical outcomes in the SGLT2i group, it is possible that a larger sample would have shown hyperglycemia-related complications.

Conclusion

Postoperative eDKA did not occur in SGLT2i patients who were instructed to stop taking their medications 5 days before open heart surgery at our institution. And, importantly, we did not observe any worse outcomes associated with this approach (albeit with more hyperglycemic events). As such, this approach of educating patients to discontinue SGLT2i use 5 days before open heart surgery may be appropriate and certainly deserves future study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. None of the authors have any official position with any government agency, and this work does not reflect positions of government, institution, or funding agencies.

Credit Authorship Contribution Statement

Jonathan S. Auerbach: Writing – review and editing, writing – original draft, supervision, methodology, investigation, conceptualization. **Ahmed Alnajar:** Methodology, formal analysis, and data curation. **Samira S. Patel:** Methodology, formal analysis, and data curation. **Hayley B.**

Gershengorn: Methodology, formal analysis, data curation, and conceptualization. **Joseph Lamelas:** Writing – review and editing, supervision, project administration, and conceptualization. **Daitiara Perez:** Writing – review and editing, investigation, and conceptualization. **Oscar D. Aljure:** Writing – review and editing. **Tanira D. Ferreir:** Writing – review and editing and resources. **Lazaro A. Gonzalez:** Writing – review and editing. **Jorge L. Cabrera:** Writing – review and editing, and project administration. All listed authors have substantially contributed to this article and the interpretation of the data. All listed authors have contributed to drafting this work and have approved the final version of this article. All authors are agreeable to be accountable for all aspects of this work and will investigate and resolve questions in regard to this.

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